1. Preliminaries

1.1. **Decimal Expansion.**

1.1.1. *Exercise:* Prove that if the decimal expansion of r is eventually periodic then r is a rational number.

Proof. Assume that the decimal expansion of r *is* eventually periodic with period p. By definition, there exists some $m \in \mathbb{N}$ such that

$$a_{n+p} = a_n \quad \text{for all } n > m$$

We can express r as follows.

$$r = \sum_{i=1}^{\infty} 10^{-i} a_i$$

We can split this sum to isolate the non-periodic prefix:

$$r = \sum_{i=1}^{m} 10^{-i} a_i + \sum_{i=m+1}^{\infty} 10^{-i} a_i$$

Since $S_1 = \sum_{i=1}^{m} 10^{-i} a_i$ is a finite sum of rationals, $S_1 \in \mathbb{Q}$.
Now let $S_2 = \sum_{i=m+1}^{\infty} 10^{-i} a_i$.

$$S_2 = \sum_{i=m+1}^{\infty} 10^{-i} a_i = \sum_{i=1}^{\infty} 10^{-(i+m)} a_{i+m}$$

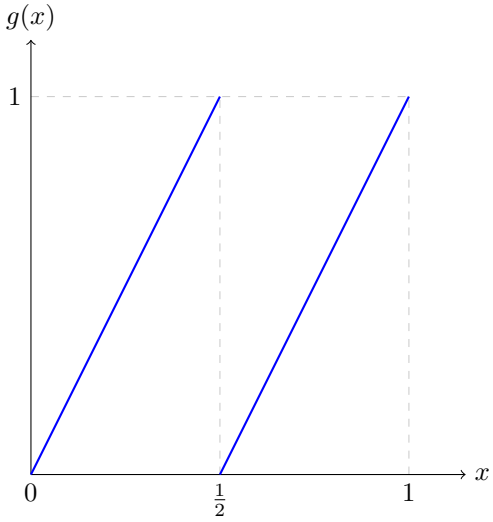
We separate the sum into periodic blocks:

$$\begin{aligned} S_2 &= \sum_{i=1}^{\infty} 10^{-(i+m)} a_{i+m} \\ &= \sum_{k=0}^{\infty} \sum_{i=1}^{p} 10^{-(i+m+kp)} a_{i+m} \\ &= \sum_{k=0}^{\infty} 10^{-pk} \sum_{i=1}^{p} 10^{-(i+m)} a_{i+m} \end{aligned}$$

Let $A = \sum_{i=1}^{p} 10^{-(i+m)} a_{i+m}$. Since A is a finite sum of rationals, $A \in \mathbb{Q}$, and $S_2 = A \cdot \sum_{k=0}^{\infty} 10^{-pk}$. We recognize this as a geometric series with common ratio 10^{-p}, and since $|10^{-p}| < 1$ for all $p \geq 1$, the series converges to a rational value. Therefore, S_2 is a product of rationals, and $r = S_1 + S_2$ is rational. □

2. Baker's transformation and the slope-2 tent map

2.0.1. *Exercise.*



[h]

2.1. **Baker's transformation and binary expansions.**